Digital Electronics (II SEM B.Tech. CSE)

Couse Code: NECE102

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Digital Electronics: Course Goals and Key Takeaways

Course Objective

To develop an understanding of digital electronics and its application which will be further useful to in the advanced courses of Electronics and Industrial Engineering.

Learning Outcomes

Upon successful completion of this course, students will:

- Gain information about the basic knowledge of digital electronics to design various types of digital circuits and systems.
- Learn how to analyse the performance and working of digital circuits.
- Develop their knowledge of digital electronics for industrial and robotic applications.

Digital Electronics: Key Topics at a Glance

Unit No.	Topics to be Covered	Lecture Hours	Learning Outcome
1.	Number System, Logic Gates, Boolean Algebra, NAND and NOR Implementation of Logic Functions, Determination of Boolean Functions from Digital Circuits.	08	Acquire an understanding of number system, basic logic gates and their applications to develop digital circuits and systems.
2.	Standard Representations of Logical Functions, SOP and POS form, Simplification of Logical Functions using K-map.	08	Acquire the knowledge to represent the logic functions in standard form and its minimization.
3.	**Combinational Logic Circuit:** Adder, Subtractor, Magnitude Comparator, Code Converters etc. **MSI Circuits:** Multiplexers, Demultiplexers, Decoders, Encoders etc.	08	Understand the design and applications of combinational logic circuits.
4.	**Sequential Circuits:** Latches and Flip-flops, Counters, Shift Registers.	10	Acquire the ability to design and perform the analysis of different sequential circuits.
5.	Analysis of Sequential circuits: State tables and state diagrams, Timing Circuits	04	Acquire an understanding on the analysis of sequential circuits.
6.	Introduction to logic families, semiconductor memories, ROM, PAL, PLA and Microprocessor	04	Develop knowledge about the various techniques for IC designing for digital circuits.

Textbook:

1. Floyd T.L., Digital Fundamentals, 10/2, Pearson Education, 2011.

Reference Books:

1. Digital Logic and Computer Design by M. Morris Mano, Prentice –Hall, 2015.
2. Fundamental of Digital Electronics by A. Anand Kumar, 4/e Prentice Hall, 2016.

Digital Electronics: Lecture Plan

Sl. No.	Topics	No of classes (tentative)
1.	Introduction to Number Systems and Conversion Techniques	8
2.	Continue…..	
3.	1’s and 2’s complement	
4.	Logic Gates (AND, OR, NOT, NAND, NOR, XOR, XNOR)	
5.	Boolean Algebra: Laws and Theorems	
6.	Boolean Function Implementation Using NAND/NOR Gates	
7.	Determination of Boolean Functions from Digital Circuits	
8.	Continue…..	
9.	SOP Standard Forms of Logical Functions	8
10.	POS Standard Forms of Logical Functions	
11.	Simplification of Logical Functions Using Boolean Algebra	
12.	Introduction to Karnaugh Maps (K-Maps)	
13.	K-Map Simplification (2-variable Functions)	
14.	K-Map Simplification (3-variable Functions)	
15.	K-Map Simplification (4-variable Functions)	
16.	Don't Care Conditions in K-Map Simplification	

Digital Electronics: Lecture Plan

17.	Introduction to Combinational Logic Circuits	8
18.	Design of Adders (Half Adder, Full Adder)	
19.	Design of Subtractors (Half Subtractor, Full Subtractor)	
20.	Magnitude Comparator Design	
21.	Code Converters (Binary to Gray, Gray to Binary, BCD to Excess-3)	
22.	Multiplexers: Design and Applications	
23.	Demultiplexers: Design and Applications	
24.	Decoders and Encoders	
25.	Introduction to Sequential Circuits	10
26.	Latches: SR Latch, D Latch	
27.	Flip-Flops: SR, JK Flip-Flops	
28.	Flip-Flops: D and T Flip-Flops	
29.	Asynchronous Counters: Design and Operation	
30.	Continue…..	
31.	Synchronous Counters: Design and Operation	
32.	Continue…..	
33.	Shift Registers: Types and Applications	
34.	Continue…..	

Digital Electronics: Weightage

Weightage of Different Components

1.	Quiz/ Assignment	10 Marks
2.	Mid Semester Examination	30 Marks
3.	Quiz/ Assignment	10 Marks
4.	End Semester Examination	50 Marks
Total Marks = 100		

Comparators

- A comparator is a logic circuit used to compare the magnitudes of two binary numbers.
- Depending on the design, it may either simply provide an output that is active (goes HIGH for example) when the two numbers are equal, or additionally provide outputs that signify which of the numbers is greater when equality does not hold.
- The X-NOR gate (coincidence gate) is a basic comparator, because its output is a 1 only if its two input bits are equal, i.e. the output is a 1 if and only if the input bits coincide.

Two binary numbers are equal, if and only if all their corresponding bits coincide. For example, two 4-bit binary numbers, $A_3A_2A_1A_0$ and $B_3B_2B_1B_0$ are equal, if and only if, $A_3 = B_3$, $A_2 = B_2$, $A_1 = B_1$ and $A_0 = B_0$. Thus, equality holds when A_3 coincides with B_3, A_2 coincides with B_2, A_1 coincides with B_1, and A_0 coincides with B_0. The implementation of this logic,

$$\text{EQUALITY} = (A_3 \odot B_3)(A_2 \odot B_2)(A_1 \odot B_1)(A_0 \odot B_0)$$

Block diagram of a 1-bit comparator.

1- Bit Magnitude Comparator

The logic for a 1-bit magnitude comparator: Let the 1-bit numbers be $A = A_0$ and $B = B_0$.

If $A_0 = 1$ and $B_0 = 0$, then $A > B$.

Therefore,

$$A > B: G = A_0\overline{B}_0$$

If $A_0 = 0$ and $B_0 = 1$, then $A < B$.

Therefore,

$$A < B: L = \overline{A}_0 B_0$$

If A_0 and B_0 coincide, i.e. $A_0 = B_0 = 0$ or if $A_0 = B_0 = 1$, then $A = B$.

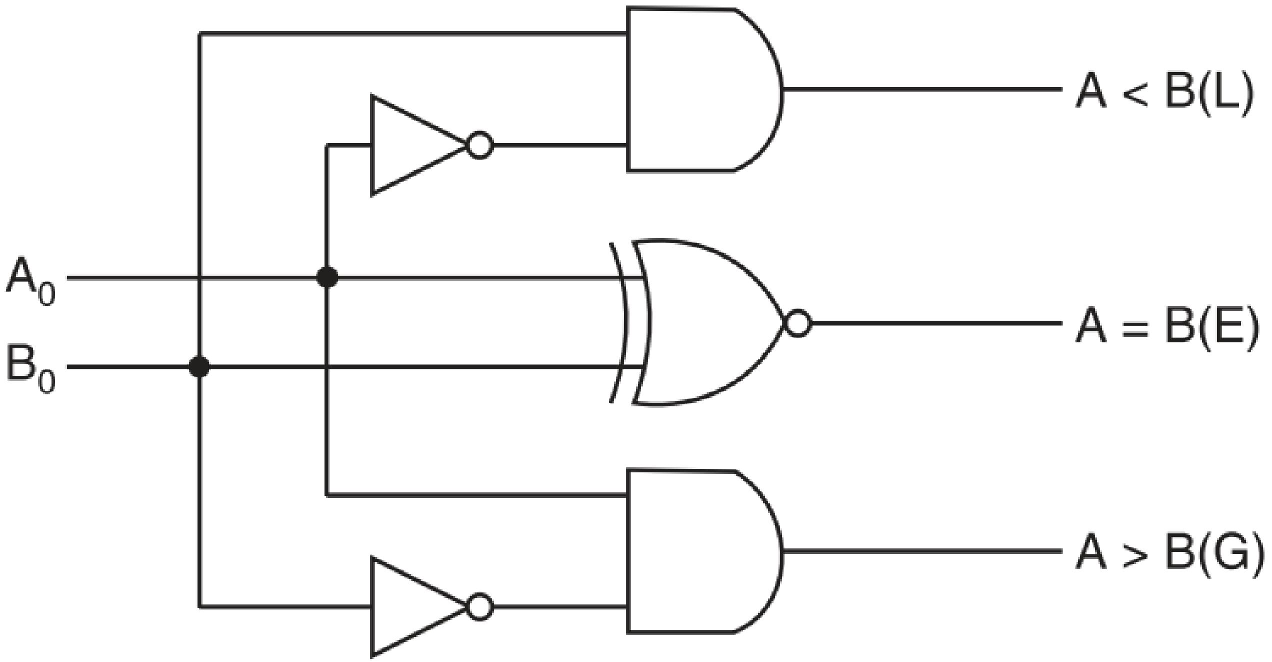
Therefore,

$$A = B : E = A_0 \odot B_0$$

1- Bit Magnitude Comparator

A_0	B_0	L	E	G
0	0	0	1	0
0	1	1	0	0
1	0	0	0	1
1	1	0	1	0

(a) Truth table



(b) Logic diagram

2- Bit Magnitude Comparator

The logic for a 2-bit magnitude comparator: Let the two 2-bit numbers be $A = A_1A_0$ and $B = B_1B_0$.

1. If $A_1 = 1$ and $B_1 = 0$, then $A > B$ or
2. If A_1 and B_1 coincide and $A_0 = 1$ and $B_0 = 0$, then $A > B$. So the logic expression for $A > B$ is

$$A > B : G = A_1\bar{B}_1 + (A_1 \odot B_1)A_0\bar{B}_0$$

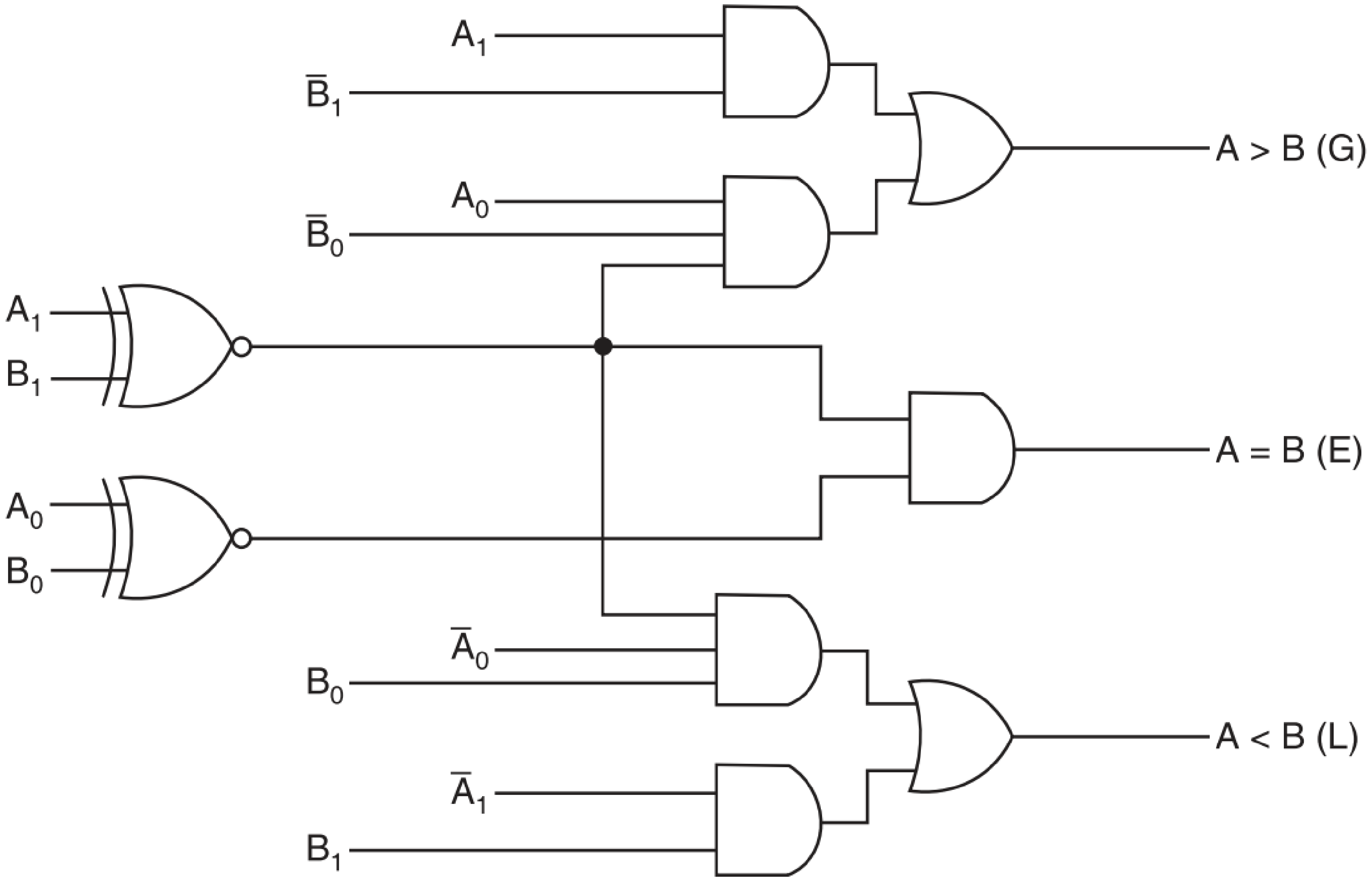
1. If $A_1 = 0$ and $B_1 = 1$, then $A < B$ or
2. If A_1 and B_1 coincide and $A_0 = 0$ and $B_0 = 1$, then $A < B$. So the expression for $A < B$ is

$$A < B : L = \bar{A}_1B_1 + (A_1 \odot B_1)\bar{A}_0B_0$$

If A_1 and B_1 coincide and if A_0 and B_0 coincide then $A = B$. So the expression for $A = B$ is

$$A = B : E = (A_1 \odot B_1)(A_0 \odot B_0)$$

2- Bit Magnitude Comparator



Logic diagram of a 2-bit magnitude comparator.

4- Bit Magnitude Comparator

The logic for a 4-bit magnitude comparator: Let the two 4-bit numbers be $A = A_3A_2A_1A_0$ and $B = B_3B_2B_1B_0$.

1. If $A_3 = 1$ and $B_3 = 0$, then $A > B$. Or
2. If A_3 and B_3 coincide, and if $A_2 = 1$ and $B_2 = 0$, then $A > B$. Or
3. If A_3 and B_3 coincide, and if A_2 and B_2 coincide, and if $A_1 = 1$ and $B_1 = 0$, then $A > B$. Or
4. If A_3 and B_3 coincide, and if A_2 and B_2 coincide, and if A_1 and B_1 coincide, and if $A_0 = 1$ and $B_0 = 0$, then $A > B$.

From these statements, we see that the logic expression for $A > B$ can be written as

$$(A > B) : G = A_3\overline{B}_3 + (A_3 \odot B_3)A_2\overline{B}_2 + (A_3 \odot B_3)(A_2 \odot B_2)A_1\overline{B}_1 + (A_3 \odot B_3)(A_2 \odot B_2)(A_1 \odot B_1)A_0\overline{B}_0$$

4- Bit Magnitude Comparator

Similarly, the logic expression for A < B can be written as

$$(A < B): L = \bar{A}_3B_3 + (A_3 \odot B_3)\bar{A}_2B_2 + (A_3 \odot B_3)(A_2 \odot B_2)\bar{A}_1B_1$$
$$+ (A_3 \odot B_3)(A_2 \odot B_2)(A_1 \odot B_1)\bar{A}_0B_0$$

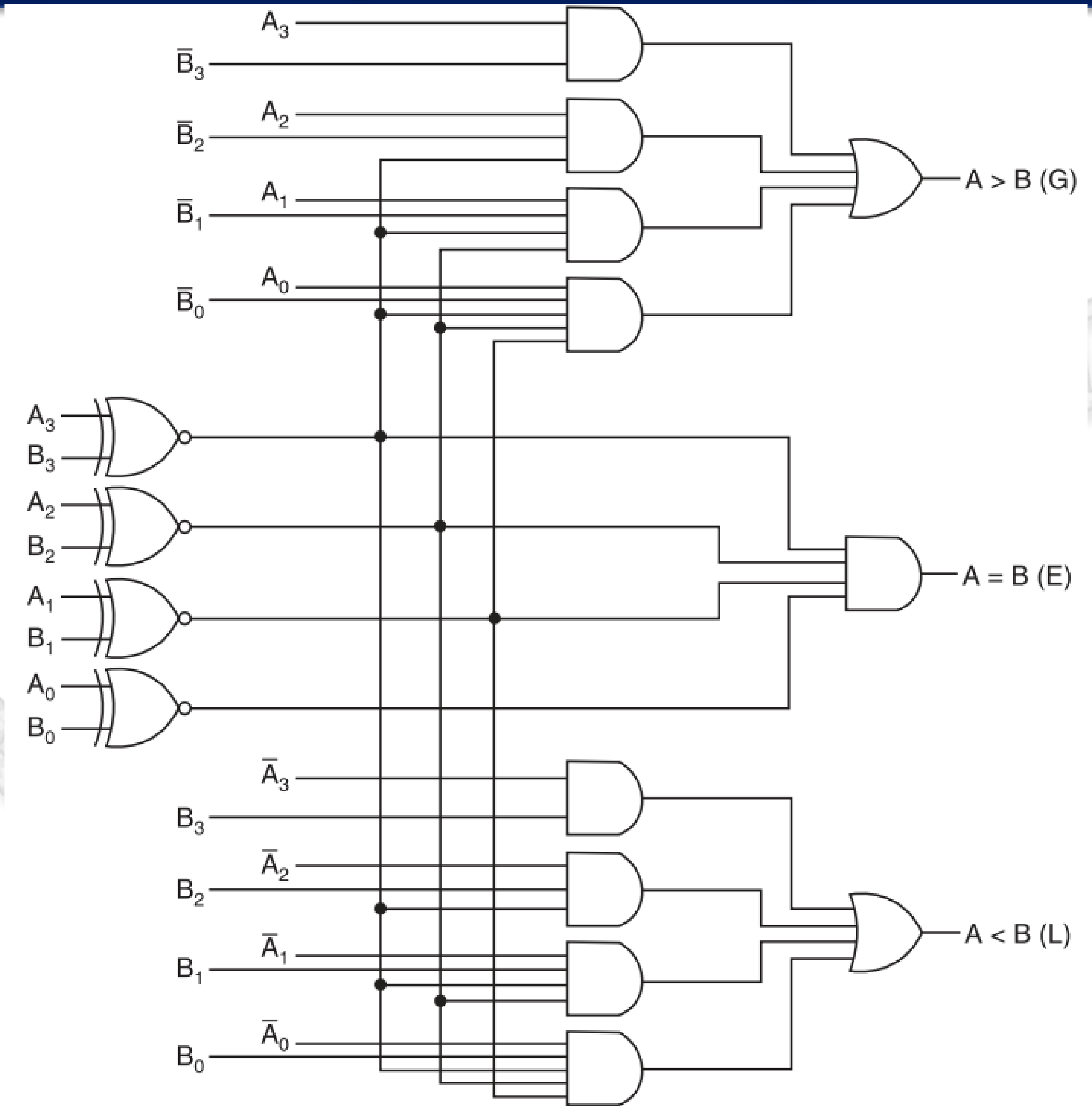
If A_3 and B_3 coincide and if A_2 and B_2 coincide and if A_1 and B_1 coincide and if A_0 and B_0 coincide, then A = B.

So the expression for A = B can be written as

$$(A = B): E = (A_3 \odot B_3)(A_2 \odot B_2)(A_1 \odot B_1)(A_0 \odot B_0)$$

Figure 7.58 shows the logic diagram of a comparator that implements the logic we have described. Note that, it provides three active-HIGH outputs: A > B, A < B, and A = B.

4- Bit Magnitude Comparator



Logic diagram of a 4-bit magnitude comparator.

8421 Binary Coded Decimal Codes

- The 8421 BCD code is so widely used that it is a common practice to refer to it simply as BCD code.
- In this code, each decimal digit, 0 through 9, is coded by a 4-bit binary number. It is also called the *natural binary code* because of the 8, 4, 2 and 1 weights attached to it.
- It is a weighted code and is also sequential. Therefore, it is useful for mathematical operations.
- The main advantage of this code is its ease of conversion to and from decimal.
- It is less efficient than the pure binary, in the sense that it requires more bits. For example, the decimal number 14 can be represented as 1110 in pure binary but as 0001 0100 in 8421 code.

DECIMAL NUMBER	BCD
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001

There are six illegal combinations 1010, 1011, 1100, 1101, 1110 and 1111 in this code, i.e. they are not part of the 8421 BCD code system.

Binary Coded Decimal Addition

- The BCD addition is performed by individually adding the corresponding digits of the decimal numbers expressed in 4-bit binary groups starting from the LSD.
- If there is no carry and the sum term is not an illegal code, no correction is needed.
- If there is a carry out of one group to the next group, or if the sum term is an illegal code, then 6 (0110) is added to the sum term of that group and the resulting carry is added to the next group. (This is done to skip the six illegal states).

- **Perform the following decimal additions in the 8421 code.**

(a)	25	$\Rightarrow$	0010	0101	(25 in BCD)
	+13		+0001	0011	(13 in BCD)
	38		0011	1000	(No carry, no illegal code. So, this is the correct sum.)

Binary Coded Decimal Addition

- **Perform the following decimal additions in the 8421 code.**

(b)	679.6	⇒		0110	0111	1001	.0110	(679.6 in BCD)
	+ 536.8			+0101	0011	0110	.1000	(536.8 in BCD)
	1216.4			1011	1010	1111	.1110	(All are illegal codes)
				+0110	+0110	+0110	+.0110	(Add 0110 to each)
				❶0001	❶0000	❶0101	❶.0100	(Propagate carry)
			+1	+1	+1	+1		
			0001	0010	0001	0110	.0100	(Corrected sum = 1216.4)
			1	2	1	6	. 4	

Binary Coded Decimal Addition

Illegal Code	Add (1) 0001	Add (2) 0010	Add (3) 0011	Add (4) 0100	Add (5) 0101	Add (6) 0110	Add (7) 0111	Add (8) 1000	Add (9) 1001
1010	1011 (Illegal Code)	1100 (Illegal Code)	1101 (Illegal Code)	1110 (Illegal Code)	1111 (Illegal Code)	0000 with a carry	0001 With a carry	0010 With a carry	0011 With a carry
1011	1100 (Illegal Code)	1101 (Illegal Code)	1110 (Illegal Code)	1111 (Illegal Code)	0000 with a carry	0001 With a carry	0010 With a carry	0011 With a carry	0100 With a carry
1100	1101 (Illegal Code)	1110 (Illegal Code)	1111 (Illegal Code)	0000 with a carry	0001 With a carry	0010 With a carry	0011 With a carry	0100 With a carry	0101 With a carry
1101	1110 (Illegal Code)	1111 (Illegal Code)	0000 with a carry	0001 With a carry	0010 With a carry	0011 With a carry	0100 With a carry	0101 With a carry	0110 With a carry
1110	1111 (Illegal Code)	0000 with a carry	0001 With a carry	0010 With a carry	0011 With a carry	0100 With a carry	0101 With a carry	0110 With a carry	0111 With a carry
1111	0000 with a carry	0001 With a carry	0010 With a carry	0011 With a carry	0100 With a carry	0101 With a carry	0110 With a carry	0111 With a carry	1000 With a carry

We add 6 in BCD because it is the smallest binary number that converts all invalid BCD sums (10–15) into valid BCD digits while generating the correct decimal carry.

The Excess Three (XS-3) Code

- The Excess-3 code, also called XS-3, is a non-weighted BCD code. This code derives its name from the fact that each binary code word is the corresponding 8421 code word plus 0011(3).
- It is a sequential code.
- The XS-3 code has six invalid states 0000, 0001, 0010, 1101, 1110 and 1111.

Decimal Digit	Excess-3 Code
0	0011
1	0100
2	0101
3	0110
4	0111
5	1000
6	1001
7	1010
8	1011
9	1100

The Gray Code

- The Gray code is a non-weighted code.
- It is not a BCD code. It is a cyclic code because successive code words in this code differ in one bit position only, i.e. it is a unit distance code.

Gray Code				Decimal	4-bit binary
1-bit	2-bit	3-bit	4-bit		
0	00	000	0000	0	0000
1	01	001	0001	1	0001
	11	011	0011	2	0010
	10	010	0010	3	0011
		110	0110	4	0100
		111	0111	5	0101
		101	0101	6	0110
		100	0100	7	0111
			1100	8	1000
			1101	9	1001
			1111	10	1010
			1110	11	1011
			1010	12	1100
			1011	13	1101
			1001	14	1110
			1000	15	1111

Code Converters

❖ A code converter is a logic circuit whose inputs are bit patterns representing numbers (or characters) in one code and whose outputs are the corresponding representations in a different code.

❖ It makes two systems compatible even though each uses a different binary code.

❖ To convert from binary code A to binary code B, the input lines must supply the bit combination of elements as specified by code A and the output lines must generate the corresponding bit combination of code B.

❖ A combinational circuit performs this transformation by means of logic gates.

Design of a 4-bit Binary-to-Gray Code Converter

4-bit binary				4-bit Gray			
B_4	B_3	B_2	B_1	G_4	G_3	G_2	G_1
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	1
0	0	1	1	0	0	1	0
0	1	0	0	0	1	1	0
0	1	0	1	0	1	1	1
0	1	1	0	0	1	0	1
0	1	1	1	0	1	0	0
1	0	0	0	1	1	0	0
1	0	0	1	1	1	0	1
1	0	1	0	1	1	1	1
1	0	1	1	1	1	1	0
1	1	0	0	1	0	1	0
1	1	0	1	1	0	1	1
1	1	1	0	1	0	0	1
1	1	1	1	1	0	0	0

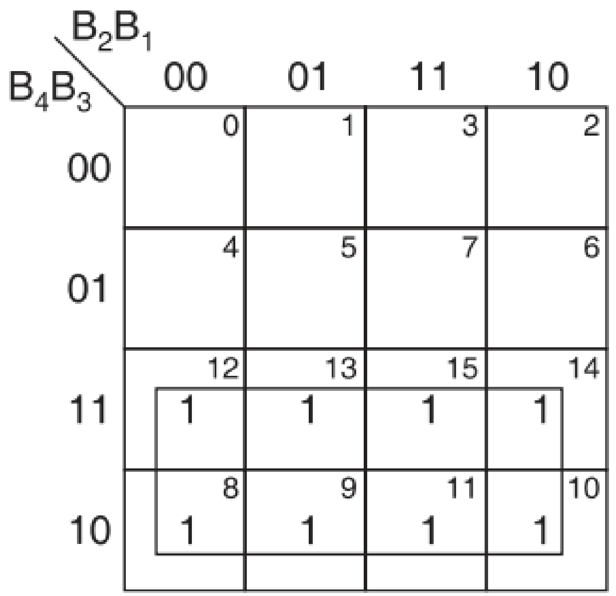
(a) Conversion table

Design of a 4-bit Binary-to-Gray Code Converter

4-bit binary				4-bit Gray			
B_4	B_3	B_2	B_1	G_4	G_3	G_2	G_1
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	1
0	0	1	1	0	0	1	0
0	1	0	0	0	1	1	0
0	1	0	1	0	1	1	1
0	1	1	0	0	1	0	1
0	1	1	1	0	1	0	0
1	0	0	0	1	1	0	0
1	0	0	1	1	1	0	1
1	0	1	0	1	1	1	1
1	0	1	1	1	1	1	0
1	1	0	0	1	0	1	0
1	1	0	1	1	0	1	1
1	1	1	0	1	0	0	1
1	1	1	1	1	0	0	0

(a) Conversion table

$$G_4 = \Sigma m(8, 9, 10, 11, 12, 13, 14, 15)$$
$$G_3 = \Sigma m(4, 5, 6, 7, 8, 9, 10, 11)$$
$$G_2 = \Sigma m(2, 3, 4, 5, 10, 11, 12, 13)$$
$$G_1 = \Sigma m(1, 2, 5, 6, 9, 10, 13, 14)$$



$G_4 = B_4$

K-map for G_4

Design of a 4-bit Binary-to-Gray Code Converter

4-bit binary				4-bit Gray			
B_4	B_3	B_2	B_1	G_4	G_3	G_2	G_1
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	1
0	0	1	1	0	0	1	0
0	1	0	0	0	1	1	0
0	1	0	1	0	1	1	1
0	1	1	0	0	1	0	1
0	1	1	1	0	1	0	0
1	0	0	0	1	1	0	0
1	0	0	1	1	1	0	1
1	0	1	0	1	1	1	1
1	0	1	1	1	1	1	0
1	1	0	0	1	0	1	0
1	1	0	1	1	0	1	1
1	1	1	0	1	0	0	1
1	1	1	1	1	0	0	0

(a) Conversion table

$$G_4 = \Sigma m(8, 9, 10, 11, 12, 13, 14, 15)$$
$$G_3 = \Sigma m(4, 5, 6, 7, 8, 9, 10, 11)$$
$$G_2 = \Sigma m(2, 3, 4, 5, 10, 11, 12, 13)$$
$$G_1 = \Sigma m(1, 2, 5, 6, 9, 10, 13, 14)$$

B_4B_3 \ B_2B_1	00	01	11	10
00	0:	1:	3:	2:
01	4: 1	5: 1	7: 1	6: 1
11	12:	13:	15:	14:
10	8: 1	9: 1	11: 1	10: 1

$G_3 = B_4 \oplus B_3$

K-map for G_3

$$G_3 = \bar{B}_4 B_3 + B_4 \bar{B}_3 = B_4 \oplus B_3$$

Design of a 4-bit Binary-to-Gray Code Converter

4-bit binary				4-bit Gray			
B_4	B_3	B_2	B_1	G_4	G_3	G_2	G_1
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	1
0	0	1	1	0	0	1	0
0	1	0	0	0	1	1	0
0	1	0	1	0	1	1	1
0	1	1	0	0	1	0	1
0	1	1	1	0	1	0	0
1	0	0	0	1	1	0	0
1	0	0	1	1	1	0	1
1	0	1	0	1	1	1	1
1	0	1	1	1	1	1	0
1	1	0	0	1	0	1	0
1	1	0	1	1	0	1	1
1	1	1	0	1	0	0	1
1	1	1	1	1	0	0	0

(a) Conversion table

$$G_4 = \Sigma m(8, 9, 10, 11, 12, 13, 14, 15)$$
$$G_3 = \Sigma m(4, 5, 6, 7, 8, 9, 10, 11)$$
$$G_2 = \Sigma m(2, 3, 4, 5, 10, 11, 12, 13)$$
$$G_1 = \Sigma m(1, 2, 5, 6, 9, 10, 13, 14)$$

B_4B_3 \ B_2B_1	00	01	11	10
00	0:	1:	3: 1	2: 1
01	4: 1	5: 1	7:	6:
11	12: 1	13: 1	15:	14:
10	8:	9:	11: 1	10: 1

$G_2 = B_3 \oplus B_2$

K-map for G_2

$$G_2 = \overline{B}_3B_2 + B_3\overline{B}_2 = B_3 \oplus B_2$$

Design of a 4-bit Binary-to-Gray Code Converter

4-bit binary				4-bit Gray			
B_4	B_3	B_2	B_1	G_4	G_3	G_2	G_1
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	1
0	0	1	1	0	0	1	0
0	1	0	0	0	1	1	0
0	1	0	1	0	1	1	1
0	1	1	0	0	1	0	1
0	1	1	1	0	1	0	0
1	0	0	0	1	1	0	0
1	0	0	1	1	1	0	1
1	0	1	0	1	1	1	1
1	0	1	1	1	1	1	0
1	1	0	0	1	0	1	0
1	1	0	1	1	0	1	1
1	1	1	0	1	0	0	1
1	1	1	1	1	0	0	0

(a) Conversion table

$$G_4 = \Sigma m(8, 9, 10, 11, 12, 13, 14, 15)$$

$$G_3 = \Sigma m(4, 5, 6, 7, 8, 9, 10, 11)$$

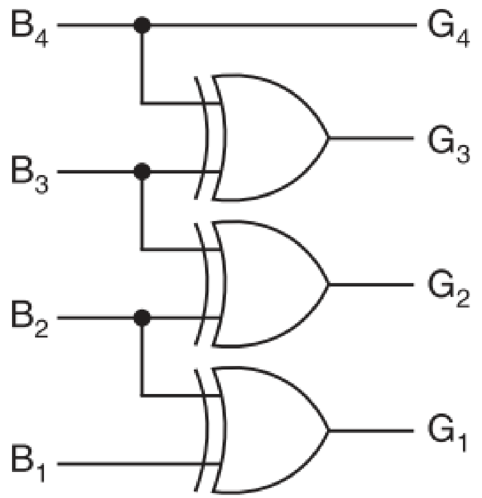
$$G_2 = \Sigma m(2, 3, 4, 5, 10, 11, 12, 13)$$

$$G_1 = \Sigma m(1, 2, 5, 6, 9, 10, 13, 14)$$

B_4B_3 \ B_2B_1	00	01	11	10
00	0	1 1	3	2 1
01	4	5 1	7	6 1
11	12	13 1	15	14 1
10	8	9 1	11	10 1

$G_1 = B_2 \oplus B_1$

K-map for G_1



(c) Logic diagram

$$G_1 = \overline{B}_2 B_1 + B_2 \overline{B}_1 = B_2 \oplus B_1$$

Design of a 4-bit Binary-to-Gray Code Converter

If an n-bit binary number is represented by $B_n\ B_{n-1} . . . B_1$ and its Gray code equivalent by $G_n\ G_{n-1} . . . G_1$, where B_n and G_n are the MSBs, then the Gray code bits are obtained from the binary code as follows.

$G_n = B_n$	$G_{n-1} = B_n \oplus B_{n-1}$	$G_{n-2} = B_{n-1} \oplus B_{n-2} \ldots$	$G_1 = B_2 \oplus B_1$

Convert the binary 1001 to the Gray code.

(a)								
	Binary	1	$\rightarrow\oplus\rightarrow$	0	$\rightarrow\oplus\rightarrow$	0	$\rightarrow\oplus\rightarrow$	1
		$\|\|$		$\|\|$		$\|\|$		$\|\|$
	Gray	1		1		0		1

Design of a 4-bit Gray-to-Binary Code Converter

4-bit Gray				4-bit binary			
G_4	G_3	G_2	G_1	B_4	B_3	B_2	B_1
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	1	0	0	1	0
0	0	1	0	0	0	1	1
0	1	1	0	0	1	0	0
0	1	1	1	0	1	0	1
0	1	0	1	0	1	1	0
0	1	0	0	0	1	1	1
1	1	0	0	1	0	0	0
1	1	0	1	1	0	0	1
1	1	1	1	1	0	1	0
1	1	1	0	1	0	1	1
1	0	1	0	1	1	0	0
1	0	1	1	1	1	0	1
1	0	0	1	1	1	1	0
1	0	0	0	1	1	1	1

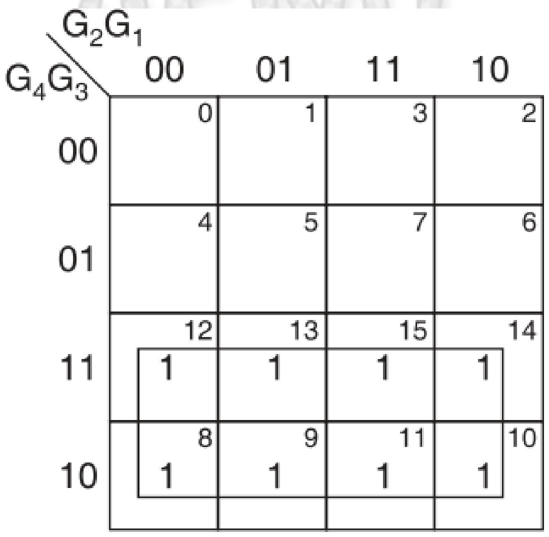
(a) Conversion table

$B_4 = \Sigma\, m(12, 13, 15, 14, 10, 11, 9, 8) = \Sigma\, m(8, 9, 10, 11, 12, 13, 14, 15)$

$B_3 = \Sigma\, m(6, 7, 5, 4, 10, 11, 9, 8) = \Sigma\, m(4, 5, 6, 7, 8, 9, 10, 11)$

$B_2 = \Sigma\, m(3, 2, 5, 4, 15, 14, 9, 8) = \Sigma\, m(2, 3, 4, 5, 8, 9, 14, 15)$

$B_1 = \Sigma\, m(1, 2, 7, 4, 13, 14, 11, 8) = \Sigma\, m(1, 2, 4, 7, 8, 11, 13, 14)$



$B_4 = G_4$

K-map for B_4

Design of a 4-bit Gray-to-Binary Code Converter

4-bit Gray				4-bit binary			
G_4	G_3	G_2	G_1	B_4	B_3	B_2	B_1
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	1	0	0	1	0
0	0	1	0	0	0	1	1
0	1	1	0	0	1	0	0
0	1	1	1	0	1	0	1
0	1	0	1	0	1	1	0
0	1	0	0	0	1	1	1
1	1	0	0	1	0	0	0
1	1	0	1	1	0	0	1
1	1	1	1	1	0	1	0
1	1	1	0	1	0	1	1
1	0	1	0	1	1	0	0
1	0	1	1	1	1	0	1
1	0	0	1	1	1	1	0
1	0	0	0	1	1	1	1

(a) Conversion table

$$B_4 = \Sigma\, m(12, 13, 15, 14, 10, 11, 9, 8) = \Sigma\, m(8, 9, 10, 11, 12, 13, 14, 15)$$
$$B_3 = \Sigma\, m(6, 7, 5, 4, 10, 11, 9, 8) = \Sigma\, m(4, 5, 6, 7, 8, 9, 10, 11)$$
$$B_2 = \Sigma\, m(3, 2, 5, 4, 15, 14, 9, 8) = \Sigma\, m(2, 3, 4, 5, 8, 9, 14, 15)$$
$$B_1 = \Sigma\, m(1, 2, 7, 4, 13, 14, 11, 8) = \Sigma\, m(1, 2, 4, 7, 8, 11, 13, 14)$$

G_4G_3 \ G_2G_1	00	01	11	10
00	0	1	3	2
01	4: 1	5: 1	7: 1	6: 1
11	12	13	15	14
10	8: 1	9: 1	11: 1	10: 1

$B_3 = G_4 \oplus G_3$

K-map for B_3

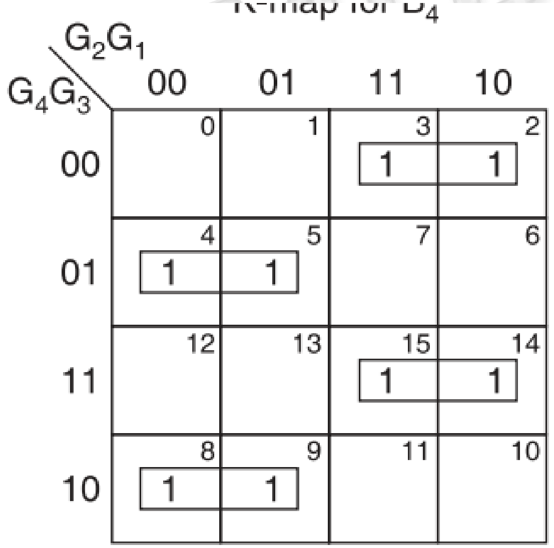
$$B_3 = \bar{G}_4G_3 + G_4\bar{G}_3 = G_4 \oplus G_3$$

Design of a 4-bit Gray-to-Binary Code Converter

4-bit Gray				4-bit binary			
G_4	G_3	G_2	G_1	B_4	B_3	B_2	B_1
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	1	0	0	1	0
0	0	1	0	0	0	1	1
0	1	1	0	0	1	0	0
0	1	1	1	0	1	0	1
0	1	0	1	0	1	1	0
0	1	0	0	0	1	1	1
1	1	0	0	1	0	0	0
1	1	0	1	1	0	0	1
1	1	1	1	1	0	1	0
1	1	1	0	1	0	1	1
1	0	1	0	1	1	0	0
1	0	1	1	1	1	0	1
1	0	0	1	1	1	1	0
1	0	0	0	1	1	1	1

(a) Conversion table

$$B_4 = \Sigma\, m(12, 13, 15, 14, 10, 11, 9, 8) = \Sigma\, m(8, 9, 10, 11, 12, 13, 14, 15)$$
$$B_3 = \Sigma\, m(6, 7, 5, 4, 10, 11, 9, 8) = \Sigma\, m(4, 5, 6, 7, 8, 9, 10, 11)$$
$$B_2 = \Sigma\, m(3, 2, 5, 4, 15, 14, 9, 8) = \Sigma\, m(2, 3, 4, 5, 8, 9, 14, 15)$$
$$B_1 = \Sigma\, m(1, 2, 7, 4, 13, 14, 11, 8) = \Sigma\, m(1, 2, 4, 7, 8, 11, 13, 14)$$



$B_2 = G_4 \oplus G_3 \oplus G_2$

K-map for B_2

$$B_2 = \bar{G}_4 G_3 \bar{G}_2 + \bar{G}_4 \bar{G}_3 G_2 + G_4 \bar{G}_3 \bar{G}_2 + G_4 G_3 G_2$$
$$= \bar{G}_4(G_3 \oplus G_2) + G_4(\overline{G_3 \oplus G_2}) = G_4 \oplus G_3 \oplus G_2 = B_3 \oplus G_2$$

Design of a 4-bit Gray-to-Binary Code Converter

4-bit Gray				4-bit binary			
G_4	G_3	G_2	G_1	B_4	B_3	B_2	B_1
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	1	0	0	1	0
0	0	1	0	0	0	1	1
0	1	1	0	0	1	0	0
0	1	1	1	0	1	0	1
0	1	0	1	0	1	1	0
0	1	0	0	0	1	1	1
1	1	0	0	1	0	0	0
1	1	0	1	1	0	0	1
1	1	1	1	1	0	1	0
1	1	1	0	1	0	1	1
1	0	1	0	1	1	0	0
1	0	1	1	1	1	0	1
1	0	0	1	1	1	1	0
1	0	0	0	1	1	1	1

(a) Conversion table

$$B_4 = \Sigma\, m(12, 13, 15, 14, 10, 11, 9, 8) = \Sigma\, m(8, 9, 10, 11, 12, 13, 14, 15)$$
$$B_3 = \Sigma\, m(6, 7, 5, 4, 10, 11, 9, 8) = \Sigma\, m(4, 5, 6, 7, 8, 9, 10, 11)$$
$$B_2 = \Sigma\, m(3, 2, 5, 4, 15, 14, 9, 8) = \Sigma\, m(2, 3, 4, 5, 8, 9, 14, 15)$$
$$B_1 = \Sigma\, m(1, 2, 7, 4, 13, 14, 11, 8) = \Sigma\, m(1, 2, 4, 7, 8, 11, 13, 14)$$

$G_4G_3 \backslash G_2G_1$	00	01	11	10
00	(0)	1 (1)	(3)	1 (2)
01	1 (4)	(5)	1 (7)	(6)
11	(12)	1 (13)	(15)	1 (14)
10	1 (8)	(9)	1 (11)	(10)

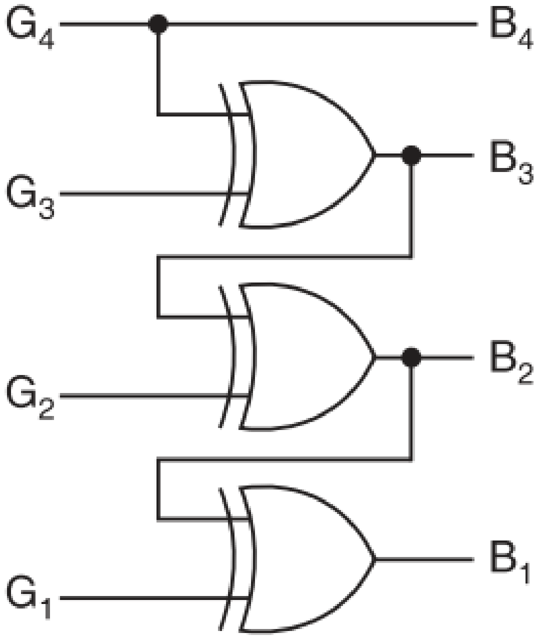
$B_1 = G_4 \oplus G_3 \oplus G_2 \oplus G_1$

K-map for B_1

maps

$$B_1 = \overline{G}_4\overline{G}_3\overline{G}_2G_1 + \overline{G}_4\overline{G}_3G_2\overline{G}_1 + \overline{G}_4G_3G_2G_1 + \overline{G}_4G_3\overline{G}_2\overline{G}_1 + G_4G_3\overline{G}_2G_1 + G_4G_3G_2\overline{G}_1 + G_4\overline{G}_3G_2G_1 + G_4\overline{G}_3\overline{G}_2\overline{G}_1$$
$$= \overline{G}_4\overline{G}_3(G_2 \oplus G_1) + G_4G_3(G_2 \oplus G_1) + \overline{G}_4G_3(\overline{G_2 \oplus G_1}) + G_4\overline{G}_3(\overline{G_2 \oplus G_1})$$
$$= (G_2 \oplus G_1)(\overline{G_4 \oplus G_3}) + (\overline{G_2 \oplus G_1})(G_4 \oplus G_3)$$
$$= G_4 \oplus G_3 \oplus G_2 \oplus G_1$$
$$= B_2 \oplus G_1$$

Design of a 4-bit Gray-to-Binary Code Converter



(c) Logic diagram

If an n-bit Gray number is represented by $G_n G_{n-1} \ldots G_1$ and its binary equivalent by $B_n B_{n-1} \ldots B_1$, then binary bits are obtained from Gray bits as follows:

$B_n = G_n$	$B_{n-1} = B_n \oplus G_{n-1}$	$B_{n-2} = B_{n-1} \oplus G_{n-2} \ldots$	$B_1 = B_2 \oplus G_1$

Convert the Gray code 1101 to binary.

The conversion is done as shown below:

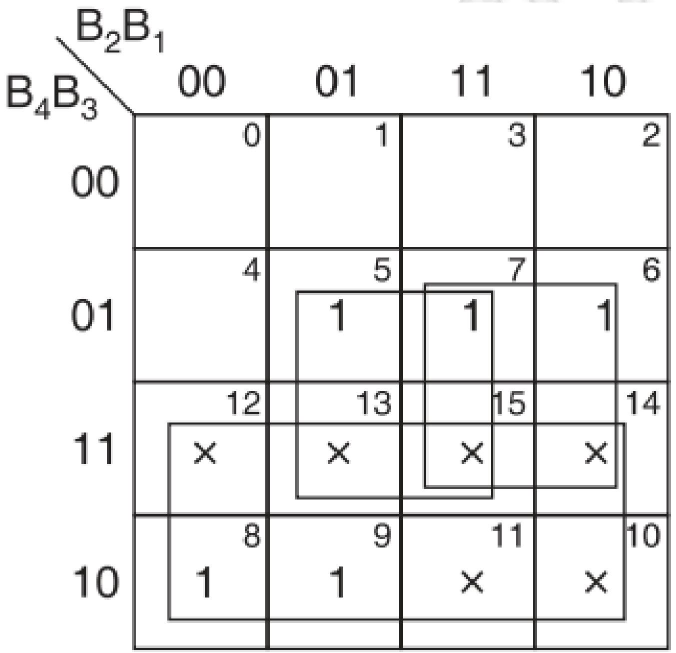
Gray	1		1		0		1
	$\Vert$	$\oplus$	$\Vert$	$\oplus$	$\Vert$	$\oplus$	$\Vert$
Binary	1		0		0		1

Design of a BCD to EX-3 Code Converter

8421 code				XS-3 code			
B_4	B_3	B_2	B_1	X_4	X_3	X_2	X_1
0	0	0	0	0	0	1	1
0	0	0	1	0	1	0	0
0	0	1	0	0	1	0	1
0	0	1	1	0	1	1	0
0	1	0	0	0	1	1	1
0	1	0	1	1	0	0	0
0	1	1	0	1	0	0	1
0	1	1	1	1	0	1	0
1	0	0	0	1	0	1	1
1	0	0	1	1	1	0	0

(a) Conversion table

$$X_4 = \Sigma\, m(5, 6, 7, 8, 9) + d(10, 11, 12, 13, 14, 15)$$
$$X_3 = \Sigma\, m(1, 2, 3, 4, 9) + d(10, 11, 12, 13, 14, 15)$$
$$X_2 = \Sigma\, m(0, 3, 4, 7, 8) + d(10, 11, 12, 13, 14, 15)$$
$$X_1 = \Sigma\, m(0, 2, 4, 6, 8) + d(10, 11, 12, 13, 14, 15)$$



$X_4 = B_4 + B_3B_2 + B_3B_1$

K-map for X_4

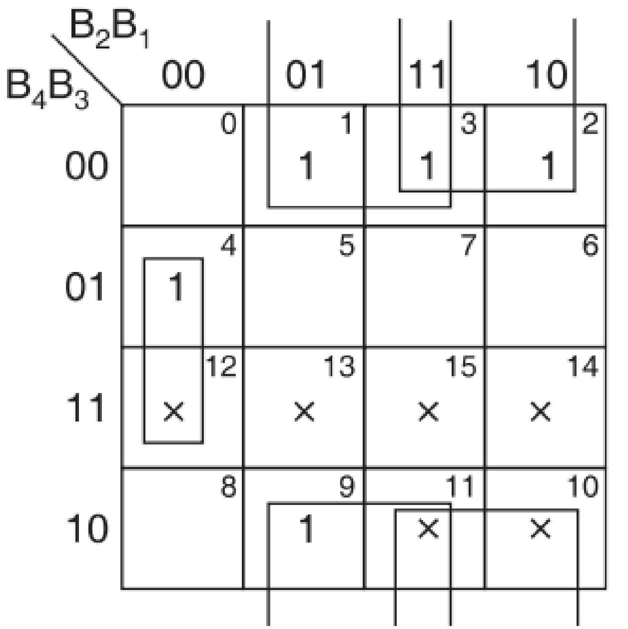
$X_4 = B_4 + B_3B_2 + B_3B_1$

Design of a BCD to EX-3 Code Converter

8421 code				XS-3 code			
B_4	B_3	B_2	B_1	X_4	X_3	X_2	X_1
0	0	0	0	0	0	1	1
0	0	0	1	0	1	0	0
0	0	1	0	0	1	0	1
0	0	1	1	0	1	1	0
0	1	0	0	0	1	1	1
0	1	0	1	1	0	0	0
0	1	1	0	1	0	0	1
0	1	1	1	1	0	1	0
1	0	0	0	1	0	1	1
1	0	0	1	1	1	0	0

(a) Conversion table

$$X_4 = \Sigma\, m(5, 6, 7, 8, 9) + d(10, 11, 12, 13, 14, 15)$$
$$X_3 = \Sigma\, m(1, 2, 3, 4, 9) + d(10, 11, 12, 13, 14, 15)$$
$$X_2 = \Sigma\, m(0, 3, 4, 7, 8) + d(10, 11, 12, 13, 14, 15)$$
$$X_1 = \Sigma\, m(0, 2, 4, 6, 8) + d(10, 11, 12, 13, 14, 15)$$



$X_3 = B_3\overline{B}_2\overline{B}_1 + \overline{B}_3B_1 + \overline{B}_3B_2$

K-map for X_3

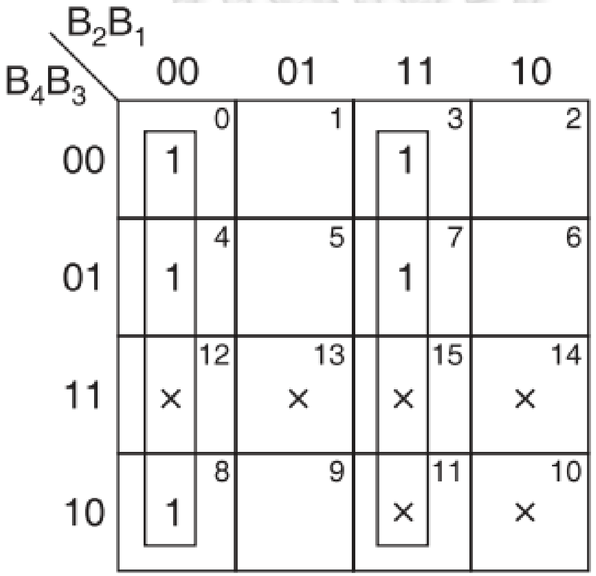
$$X_3 = B_3\overline{B}_2\overline{B}_1 + \overline{B}_3B_1 + \overline{B}_3B_2$$

Design of a BCD to EX-3 Code Converter

8421 code				XS-3 code			
B_4	B_3	B_2	B_1	X_4	X_3	X_2	X_1
0	0	0	0	0	0	1	1
0	0	0	1	0	1	0	0
0	0	1	0	0	1	0	1
0	0	1	1	0	1	1	0
0	1	0	0	0	1	1	1
0	1	0	1	1	0	0	0
0	1	1	0	1	0	0	1
0	1	1	1	1	0	1	0
1	0	0	0	1	0	1	1
1	0	0	1	1	1	0	0

(a) Conversion table

$$X_4 = \Sigma\, m(5, 6, 7, 8, 9) + d(10, 11, 12, 13, 14, 15)$$
$$X_3 = \Sigma\, m(1, 2, 3, 4, 9) + d(10, 11, 12, 13, 14, 15)$$
$$X_2 = \Sigma\, m(0, 3, 4, 7, 8) + d(10, 11, 12, 13, 14, 15)$$
$$X_1 = \Sigma\, m(0, 2, 4, 6, 8) + d(10, 11, 12, 13, 14, 15)$$



$X_2 = \overline{B}_2\overline{B}_1 + B_2B_1$

K-map for X_2

$$X_2 = \overline{B}_2\overline{B}_1 + B_2B_1$$

Design of a BCD to EX-3 Code Converter

8421 code				XS-3 code			
B_4	B_3	B_2	B_1	X_4	X_3	X_2	X_1
0	0	0	0	0	0	1	1
0	0	0	1	0	1	0	0
0	0	1	0	0	1	0	1
0	0	1	1	0	1	1	0
0	1	0	0	0	1	1	1
0	1	0	1	1	0	0	0
0	1	1	0	1	0	0	1
0	1	1	1	1	0	1	0
1	0	0	0	1	0	1	1
1	0	0	1	1	1	0	0

(a) Conversion table

$$X_4 = \Sigma\, m(5, 6, 7, 8, 9) + d(10, 11, 12, 13, 14, 15)$$
$$X_3 = \Sigma\, m(1, 2, 3, 4, 9) + d(10, 11, 12, 13, 14, 15)$$
$$X_2 = \Sigma\, m(0, 3, 4, 7, 8) + d(10, 11, 12, 13, 14, 15)$$
$$X_1 = \Sigma\, m(0, 2, 4, 6, 8) + d(10, 11, 12, 13, 14, 15)$$

B_4B_3 \ B_2B_1	00	01	11	10
00	[0] 1	[1]	[3]	[2] 1
01	[4] 1	[5]	[7]	[6] 1
11	[12] ×	[13] ×	[15] ×	[14] ×
10	[8] 1	[9]	[11] ×	[10] ×

$X_1 = \overline{B}_1$

K-map for X_1

$$X_1 = \overline{B}_1$$